

Derek Wang

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EDUCATION

University of Waterloo

Bachelor of Applied Science in Computer Engineering (Co-op)

Expected Apr 2030

GPA: 3.7

EXPERIENCE

Backend Software Engineer

Kenosis Technologies Inc.

March 2026 – Present

Mississauga, ON, Canada

- Built backend APIs and microservices for an AI-powered fraud detection platform, reduced API latency by 35%
- Designed schemas and ingestion pipelines for job posting data, improving update frequency from daily to hourly
- Integrated LLM inference into backend services via async queues, reducing time-to-flag suspicious postings by 60%
- Developed candidate distance and map data APIs, enabling employers to filter and visualize applicants by location
- Authored internal API guides and onboarding runbooks; presented architecture decisions to cross-functional stakeholders to drive adoption of new tooling

Autonomous Self Driving Software Engineer

WATonomous

Sept 2025 – Present

Waterloo, ON, Canada

- Architected and deployed ROS2 navigation stack (costmaps, A* planning, Pure Pursuit) enabling autonomous rover path following in dynamic environments (hills, obstacles, holes), simulating Mars environment
- Led YOLO11s model retraining POC, improving detection mean average precision from 30% to 80% across 3,448 labeled images via hyperparameter tuning and augmentation redesign
- Optimized YOLO11s inference to 30 FPS on rover hardware via ONNX export and onnxslim optimization

Autonomy Developer

Waterloo Aerial Robotics Group

Aug 2025 – Feb 2026

Waterloo, ON, Canada

- Developed Python hardware-in-the-loop UAV simulation using Ardupilot Mission Planner, Pixhawk flight controller and MAVLink protocol for the 2026 UAS Student Competition
- Eliminated GPS teleportation bugs upon injection and achieved smooth trajectory interpolation at 24 FPS via periodic position updates, drone speed control, position thresholding, and trajectory smoothing algorithms

PROJECTS

Reparo - AI Repair Tool (Hack Canada 2026 Winner) | *Python, FastAPI, Swift, Gemini, SerpAPI, Reactiv*

- Generated 5+ structured repair outputs from a single device image using a computer-vision + Gemini reasoning pipeline that performs component detection and contextual repair inference for consumer devices
- Built a FastAPI orchestration service that normalizes unstructured LLM responses into a typed repair schema, enabling deterministic UI rendering of repair steps, cost estimates, and required tools
- Designed a multi-stage parts retrieval pipeline (SerpAPI → Shopify storefront discovery → Shopify product APIs) to automatically source purchasable repair components surfaced in a Swift Reactiv App Clip

Silhouette - Cursor for Fashion | *Node.js, Next.js, Express, MongoDB, Gemini, Overshoot.ai SDK*

- Delivered personalized fashion recommendations in under 2s at by building Gemini pipelines with MERN stack that orchestrate async inference, product recommendations, and style constraints over Next.js live camera input
- Secured a VC referral to continue development as a Proof of Concept (Tailor AI) for Merian Brothers

Github Repository Analyzer | *Node.js, Express, React, Gemini, Octokit, Puppeteer*

- Designed and Shipped a full-stack AI-powered GitHub repository analyzer that reduced codebase evaluation time by 98% to under 90 seconds per repository, making technical repos accessible to non-technical stakeholders
- Integrated Octokit and Puppeteer to automatically extract repo metadata, file trees, and source code for analysis
- Developed a 3-agent Gemini AI pipeline generating 3 different audience-specific reports from a single codebase scan

Autonomous Vehicle Control System | *C++, ROS2, Docker, CMake, Foxglove*

- Built a C++ LiDAR perception pipeline generating real-time occupancy costmaps at 0.1m resolution in Foxglove
- Developed ROS2 autonomous navigation stack with Docker, validated 100% obstacle avoidance in 50+ simulations

TECHNICAL SKILLS

Languages: C++, Java, Python, JavaScript, TypeScript, SQL

Frameworks: React, Next.js, Node.js, Express, FastAPI, Vite

Cloud/DevOps: AWS, Azure, Docker, GitHub Actions

Libraries/Tools: PyTorch, OpenCV, SQLite, PostgreSQL, MongoDB, Git, ROS2, Foxglove, Fusion 360